MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3670 PRIMARY CYANIDE DESTRUCTION
DESIGN CALCULATION OF 3670-TNK-001

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

This document is to Prepare for design of **Primary Cyanide Destruction (3670)-** Tank Foundation. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2.EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in Excel and Staad. For member strength calculation force values are directly taken from Load Data Provided & Output of Staad analysis.

4. DESIGN REQUIREMENTS

4.1 Design Loads

All Design Loads are indicated in Foundation and Load Plan (Doc. No. MD-216-3670-EG-ME-PLN- 1001) attached in Attachment 1. Also indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

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5. CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M and API 650.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

5.3 Foundations for Aboveground Tanks

Foundation shall be Raft Foundation with a Concrete Ringwall.

5.4 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Column	50
	Raft	50

5.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 450 kN/m^2 at 1.2m depth below EGL (ie. RL 911.00m) refer Table 5 - BH-27 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

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6. DESIGN CALCULATION FOR FOUNDATION

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DESIGN OF CYANIDE DESTRUCTION PUMP TANK (3670-TNK-001)

	Along-X	Along-Z
Size of column :	0.70m	0.70m
Size of fdn. raft :	1.75m	2.50m
Thickness of raft :	0.40 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.20 m	
U/S level Base Plate :	1.40 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

SYMBOLS AND NOTATIONS

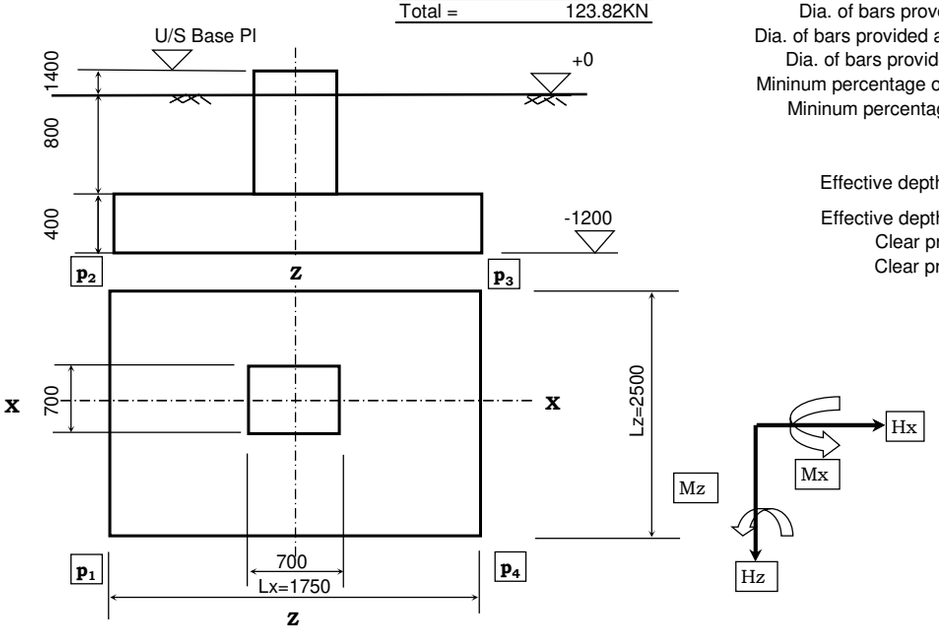
Dead Load	D
Seismic Load	E
Wind Load	W
Live Load	L
Fluid Load	F

Calculation of Weight of Foundation :

Weight of raft & Col.= (1.75*2.5*0.4*24)+(0.7*0.7*2.2*24) = 67.87KN

Weight of soil = ((1.75*2.5)-(0.7*0.7))*0.8*18 = 55.94KN

Total = 123.82KN



Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	10mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	10mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 317mm = 0.317m

Effective depth of slab along Z direction = 301mm = 0.301m

Clear projection along Z direction = 0.9m

Clear projection along X direction = 0.525m

H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Dead Load (D)	26.72	0.01	0.01	0.06	0.01
2	Seismic Load (E)	7.98	6.44	6.44	2.94	2.52
3	Wind Load (W)	11.10	8.30	8.30	4.10	3.90
4	Live Load (L)	2.78	0.00	0.00	0.01	0.00
5	Fluid Load (F)	203.45	0.07	0.07	0.24	0.03

Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	D + F	230.2	0.1	0.3	0.1	0.0	353.99	0.08	0.51	0.08	0.25
202	D + F + L	233.0	0.1	0.3	0.1	0.0	356.77	0.08	0.52	0.08	0.25
203	D + F + W	241.3	8.4	4.4	8.4	3.9	365.09	8.38	26.19	8.38	25.73
204	0.6 D + W	27.1	8.3	4.1	8.3	3.9	150.95	8.31	25.73	8.31	25.50
205	0.6 D + 0.7 E	21.6	4.5	2.1	4.5	1.8	95.91	4.51	13.83	4.51	13.51
Ultimate Conditions											
301	1.4 D + 1. 4F	322.2	0.1	0.4	0.1	0.1	495.59	0.11	0.71	0.11	0.35
302	1.2 D + 1.2 F + 1.6 L	280.7	0.1	0.4	0.1	0.0	454.00	0.10	0.63	0.10	0.30
303	1.2 D + 1.0 E + 0.5 L	41.4	6.5	3.0	6.5	2.5	190.02	6.45	19.79	6.45	19.31
304	0.9 D + 1.0 E	32.0	6.4	3.0	6.4	2.5	143.47	6.45	19.76	6.45	19.30
305	0.9 D + 1.6 W	41.8	13.3	6.6	13.3	6.2	153.25	13.29	41.17	13.29	40.80
306	1.2 D + 1.6 W + 0.5 L	51.2	13.3	6.6	13.3	6.3	199.80	13.29	41.20	13.29	40.81

Check for overturning and sliding:

	Load Cases:	Load at b/o fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
201	D + F	353.99	0.08	0.51	0.08	0.25	609.73	1784.22	1769.95	1769.95
202	D + F + L	356.77	0.08	0.52	0.08	0.25	602.65	1798.24	1783.85	1783.85
203	D + F + W	365.09	8.38	26.19	8.38	25.73	12.20	17.74	17.43	17.43
204	0.6 D + W	150.95	8.31	25.73	8.31	25.50	5.13	7.40	7.27	7.27
205	0.6 D + 0.7 E	95.91	4.51	13.83	4.51	13.51	6.07	8.88	8.50	8.50

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Load Combination & Calculation of Base pressure

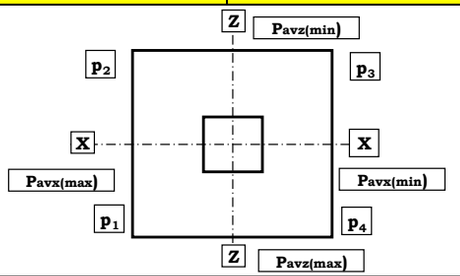
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 20 kN/m²

Net SBC of soil for Dead Load = 450 kN/m² For Seismic load case = 562.5 kN/m²

Gross SBC of soil for Dead Load = 471.6 kN/m² For Seismic load case = 584.1 kN/m²



Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m²)												
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)².Lz	6.Mx/ Lx.(Lz)²	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
201	D + F	353.99	0.51	0.25	80.91	0.40	0.14	81.45	81.17	80.65	80.38	59.85	81.31	80.52	81.05	80.78	O.K
202	D + F + L	356.77	0.52	0.25	81.55	0.41	0.14	82.09	81.82	81.28	81.01	60.49	81.96	81.15	81.69	81.42	O.K
203	D + F + W	365.09	26.19	25.73	83.45	20.52	14.11	118.09	89.86	77.04	48.81	96.49	103.98	62.93	97.57	69.34	O.K
204	0.6 D + W	150.95	25.73	25.50	34.50	20.17	13.99	68.66	40.68	28.33	0.35	47.06	54.67	14.34	48.50	20.52	O.K
205	0.6 D + 0.7 E	95.91	13.83	13.51	21.92	10.84	7.41	40.17	25.35	18.49	3.67	18.57	32.76	11.08	29.33	14.51	O.K
301	1.4 D + 1. 4F	495.59	0.71	0.35	113.28	0.56	0.19	114.02	113.64	112.91	112.53	92.42	113.83	112.72	113.47	113.09	
302	1.2 D + 1.2 F + 1.6 L	454.00	0.63	0.30	103.77	0.49	0.16	104.42	104.10	103.44	103.12	82.82	104.26	103.28	103.93	103.61	
303	1.2 D + 1.0 E + 0.5 L	190.02	19.79	19.31	43.43	15.51	10.59	69.53	48.35	38.51	17.33	47.93	58.94	27.92	54.02	32.84	
304	0.9 D + 1.0 E	143.47	19.76	19.30	32.79	15.49	10.59	58.86	37.69	27.89	6.72	37.26	48.28	17.31	43.38	22.21	
305	0.9 D + 1.6 W	153.25	41.17	40.80	35.03	32.26	22.38	89.67	44.91	25.15	-19.61	68.07	67.29	2.77	57.41	12.65	
306	1.2 D + 1.6 W + 0.5 L	199.80	41.20	40.81	45.67	32.28	22.39	100.34	55.56	35.77	-9.00	78.74	77.95	13.39	68.06	23.28	
Max :		496	41	41													
Min :		96	1	0													

Max pressure at base of foundation without surcharge excluding weight of soil (Dead load cases) = 59.85 kN/m2

Max pressure at base of foundation including surcharge excluding weight of soil (Dead load cases) = 79.85 kN/m2

Max pressure at base of foundation without surcharge excluding weight of soil (Dead+Temperature+Seismic load cases) = 96.49 kN/m2

Max pressure at base of foundation including surcharge excluding weight of soil (Dead+ Temperature+Seismic load cases) = 116.49 kN/m2

Sample calculation of values of base pressure:

Load case : D + F + L

Maximum pressure, p_1 (pmax) = 81.55+0.41+0.14= 82.09 kN/m²

p_2= 81.55+0.41-0.14= 81.82 kN/m²

p_3= 81.55-(0.41+0.14)= 81.01 kN/m²

p_4= 81.55-0.41+0.14= 81.28 kN/m²

Maxm pressure excluding weight of soil (p_{max}-γh)= 82.09-1.2*18= 60.49 kN/m²

Maximum average Pressure along X, $p_{avx(max)}$ = (82.09+81.82)/2= 81.95 kN/m2

Minimum average Pressure along X, $p_{avx(minx)}$ = (81.01+81.28)/2= 81.14 kN/m2

Maximum average Pressure along Z, $p_{avz(max)}$ = (82.09+81.28)/2= 81.68 kN/m2

Minimum average Pressure along Z, $p_{avz(minx)}$ = (81.01+81.82)/2= 81.41 kN/m2

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Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.7+2*0.5*0.301)+(0.7+2*0.5*0.301))*2

= 4.004 m

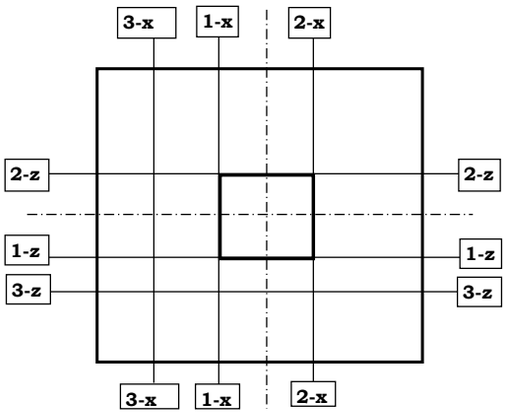
Punching Area= ((0.7+2*0.5*0.301)*(0.7+2*0.5*0.301))

= 1 m2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*24+0.8*18)

= 24.00 kN/m2

Area of footing, Af= 4.38 m2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	B.M_{1x} kNm	B.M_{2x} kNm	S.F_{3x} kN	S.F(punching) kN
Ultimate Conditions								
301	1.4 D + 1. 4F	113.50	113.05	113.70	12.36	12.24	18.67	301.31
302	1.2 D + 1.2 F + 1.6 L	103.97	103.57	104.14	11.05	10.94	16.68	269.23
303	1.2 D + 1.0 E + 0.5 L	49.63	37.23	55.25	4.39	0.97	6.88	65.59
304	0.9 D + 1.0 E	38.99	26.60	44.60	2.92	-0.50	4.67	29.67
305	0.9 D + 1.6 W	47.93	22.13	59.62	5.08	-2.04	8.21	37.22
306	1.2 D + 1.6 W + 0.5 L	58.58	32.76	70.28	6.55	-0.57	10.42	73.13

Sample calculation of BM & SF

Load case : 301

p_{1x} = 113.83-(113.83-112.72)*0.525/1.75 = 113.50 kN/m^2

p_{2x} = 112.72+(113.83-112.72)*0.525/1.75 = 113.05 kN/m^2

p_{3x} = 113.83-(113.83-112.72)*(0.525-0.317)/1.75 = 113.70 kN/m^2

Overhang = 0.525 m

Effective depth = 0.317 m

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BM_{1x} = (113.50-24)*0.525*0.525/2+0.5*(113.83-113.50)*0.525*0.525*0.67 = 12.36 kN-m/m

BM_{2x} = (113.05-24)*0.525*0.525/2-0.5*(-112.72+113.05)*0.525*0.525*0.67 = 12.24 kN-m/m

SF_{3x} = (113.70-24)*(0.525-0.317)+0.5*(113.83-113.70)*(0.525-0.317) = 18.67 kN/m

$S.F_{(punching)}$ = (4.375-1)*(113.28-24) = 301.31 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	12.36	0.12	0.16	0.000	124.17	570.60	16ϕT	352.37	300	200	1005.31
At top	-2.04	-0.02	0.03	0.000	20.40	190.20	10ϕT	412.93	300	200	392.70

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, $(\phi\ Vc)/bd$ = 0.740 N/mm2

Shear at a distance 'd' from face of column, Vu= 18.67 kN/m

Shear stress , τ_v= (18.67*1000)/(1000*317)

= 0.06 N/mm2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 301.31 kN

Strength of Concrete in shear Vc1 = 3754.55 kN Table 22.6.5.2_ (a)

Strength of Concrete in shear Vc2 = 2242.35 kN Table 22.6.5.2_ (b)

Strength of Concrete in shear Vc3 = 2503.03 kN Table 22.6.5.2_ (c)

Design Vc= 2242.35 kN

Allowable shear stress, $(\phi\ Vc)/bd$ = 1.32 N/mm2

Design punching shear stress, τ = (301.31*1000)/(4004*317)

= 0.24 N/mm2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p_{1z} =pressure at the face of column at section 1z-1z in kN/m2.
p_{2z} =pressure at the face of column at section 2z-2z in kN/m2.
p_{3z} =pressure at a distance d from the face of column at section 3z-3z in kN/m2.
$B.M_{1z}$ =Moment at the face of column at section 1z-1z in kN-m/m.
$B.M_{2z}$ =Moment at the face of column at section 2z-2z in kN-m/m.
$S.F_{3z}$ = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
301	1.4 D + 1. 4F	113.33	113.23	113.38	36.22	36.10	53.57
302	1.2 D + 1.2 F + 1.6 L	103.81	103.73	103.85	32.36	32.26	47.86
303	1.2 D + 1.0 E + 0.5 L	46.40	40.46	48.95	11.14	4.60	16.46
304	0.9 D + 1.0 E	35.76	29.83	38.31	6.83	0.29	10.09
305	0.9 D + 1.6 W	41.30	28.76	46.69	11.38	-2.44	16.80
306	1.2 D + 1.6 W + 0.5 L	51.94	39.40	57.33	15.69	1.86	23.18

Sample calculation of BM & SF

Load case : 301

p_{1z} = 113.47-(113.47-113.09)*0.9/2.5= 113.33 kN/m2

p_{2z} = 113.09+(113.47-113.09)*0.9/2.5= 113.23 kN/m2 Overhang = 0.9 m

p_{3z} = 113.47-(113.47-113.09)*(0.9-0.301)/2.5= 113.38 kN/m2 Effective depth = 0.301 m

BM_{1z} = (113.33-24)*0.9*0.9/2+0.5*(113.47-113.33)*0.9*0.9*0.67 = 36.22 kN-m/m

BM_{2z} = (113.23-24)*0.9*0.9/2-0.5*(-113.09+113.23)*0.9*0.9*0.67 = 36.10 kN-m/m

SF_{3z} = (113.38-24)*(0.9-0.301)+0.5*(113.47-113.38)*(0.9-0.301) = 53.57 kN/m

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Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm2)	ρ=	Ast (mm2)	Minm. Ast (mm2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm2)
At bottom	36.22	0.400	0.533	0.0013	405.95	541.80	16ϕT	371.10	300	200	1005.3
At top	-2.44	-0.027	-0.036	-0.0001	-0.26	180.60	10ϕT	434.88	300	200	392.7

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.74 N/mm^2

Shear at a distance 'd' from face of column, V_u= 53.57 kN/m

Shear stress , τ_v= (53.57*1000)/(1000*301)

= 0.18 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	16ϕT	200	c/c
	At Top	10ϕT	200	c/c
Along Z Direction :				
	At Bottom	16ϕT	200	c/c
	At Top	10ϕT	200	c/c

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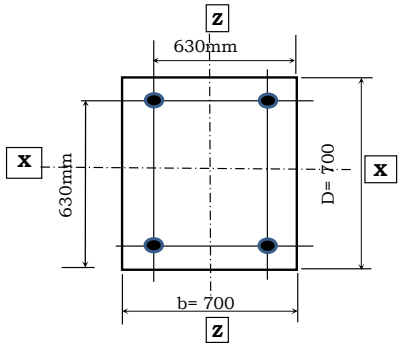
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DESIGN OF COLUMN MKD C1

Input:

Clear cover to reinf.=	50mm
Dia. of main reinf.=	20mm
Dia. of link=	10mm
Effective cover =	70mm
Grade of Concrete (f_c') =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	0.50%
Max. Steel Ratio	4.00%

Table 20.6.1.3.1 of SBC-304



For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2 of SBC-304

Unsupproted length of column about X (l_{ux}) =	2200mm
Unsupproted length of column about Z (l_{uz}) =	2200mm

For Columns not braced against sideway, Where, Effective length factor K= 2

Slenderness ratio, $K l_{ux} / r_x$ = 20.95 < 22 SECTION 6.2.5a Radius of gyration r_x= 210 mm

$K l_{uz} / r_z$ = 20.95 < 22 SECTION 6.2.5a Radius of gyration r_z= 210 mm

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Dead Load (D)	26.72	0.01	0.01	0.06	0.01
2	Seismic Load (E)	7.98	6.44	6.44	2.94	2.52
3	Wind Load (W)	11.10	8.30	8.30	4.10	3.90
4	Live Load (L)	2.78	0.00	0.00	0.01	0.00
5	Fluid Load (F)	203.45	0.07	0.07	0.24	0.03
			γ_x =	0.83		
			γ_z =	0.83		

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Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = Pn/f'cbD (N/mm2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
301	1.4 D + 1.4 F	32.22	0.07	0.03	49.58	0.10	0.05	0.03
302	1.2 D + 1.2 F + 1.6 L	28.07	0.06	0.03	43.18	0.09	0.04	0.03
303	1.2 D + 1.0 E + 0.5 L	4.14	1.72	1.67	6.37	2.65	2.57	0.00
304	0.9 D + 1.0 E	3.20	1.72	1.67	4.93	2.64	2.57	0.00
305	0.9 D + 1.6 W	4.18	3.65	3.57	6.43	5.62	5.50	0.00
306	1.2 D + 1.6 W + 0.5 L	5.12	3.59	3.55	7.88	5.52	5.46	0.00

Reinforcement: Provide,

4-Nos	20 mm Dia	1257.14 Sq mm	(At Corner)
4-Nos	20 mm Dia	1257.14 Sq mm	(Along Z axis)
4-Nos	20 mm Dia	1257.14 Sq mm	(Along Y axis)

Asc pvd. = 3771.43 Sq mm

Total Asc = 0.8 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 1604.93 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_cb^2D$	$R_{nx}=M_{nox}/f'_cb^2D$						
301	0.05	0.05	60.0	60.0	0.55	1.16	0.00	O.K
302	0.05	0.05	60.0	60.0	0.55	1.16	0.00	O.K
303	0.05	0.05	60.0	60.0	0.55	1.16	0.05	O.K
304	0.05	0.05	60.0	60.0	0.55	1.16	0.05	O.K
305	0.05	0.05	60.0	60.0	0.55	1.16	0.13	O.K
306	0.05	0.05	60.0	60.0	0.55	1.16	0.12	O.K

Links, 10mm dia @ 300 C/C

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

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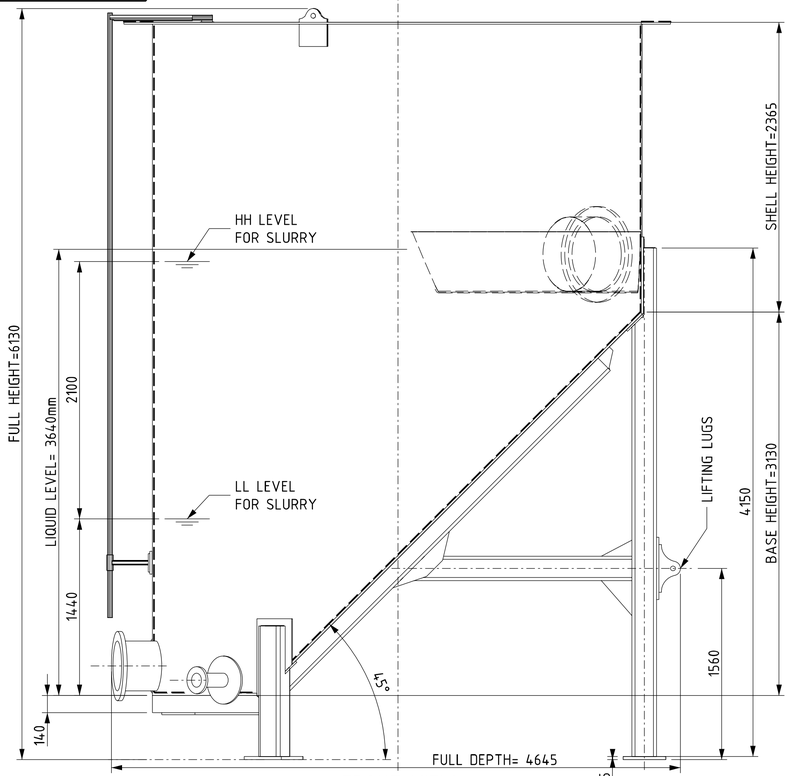
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ATTACHMENT 1 LOAD DATA DRAWINGS

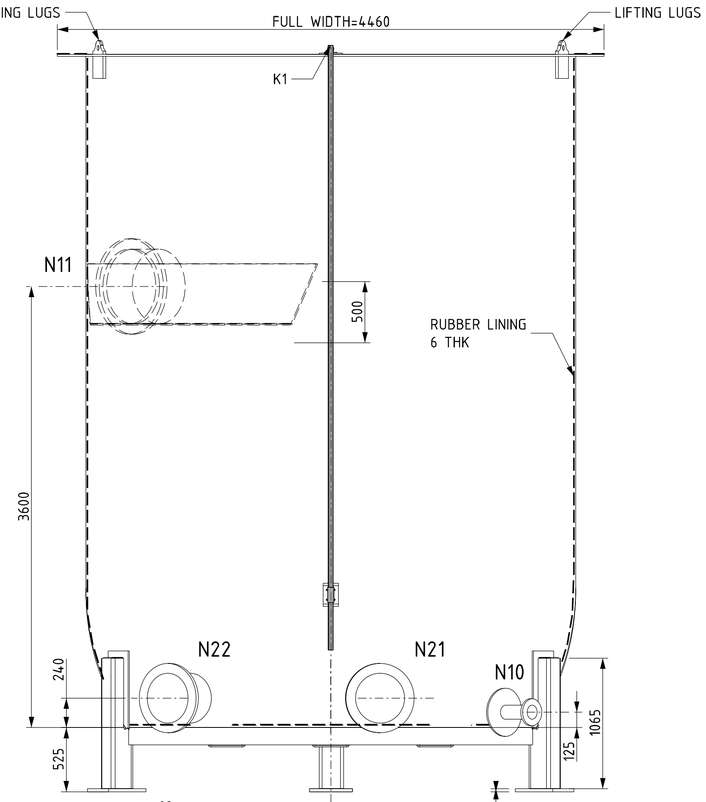
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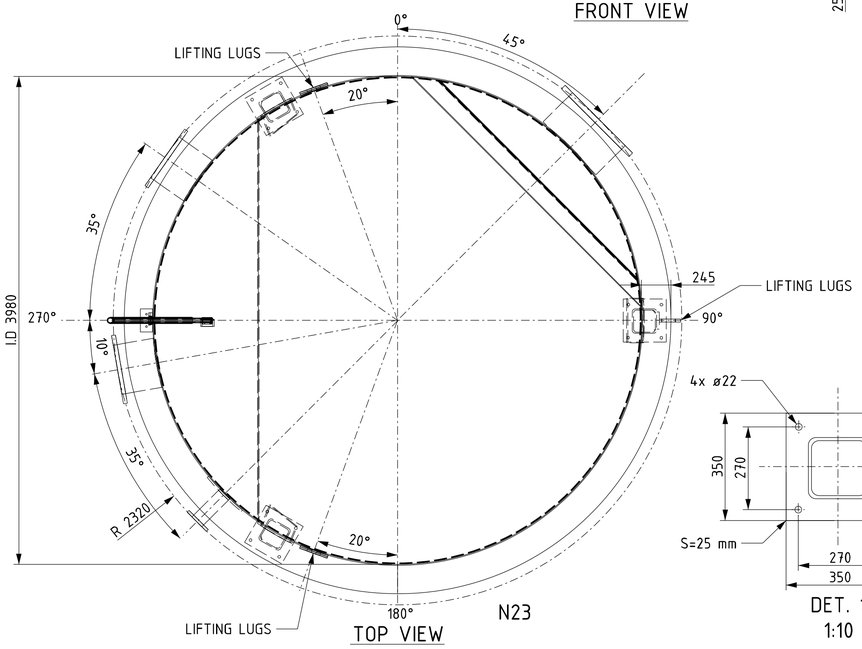
DRAWING NUMBER MD-216-3670-EG-ME-GAD-1007 | REV A01



FRONT VIEW

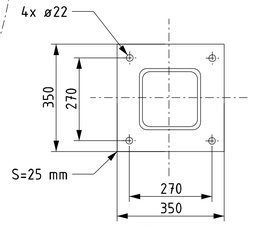


LEFT VIEW

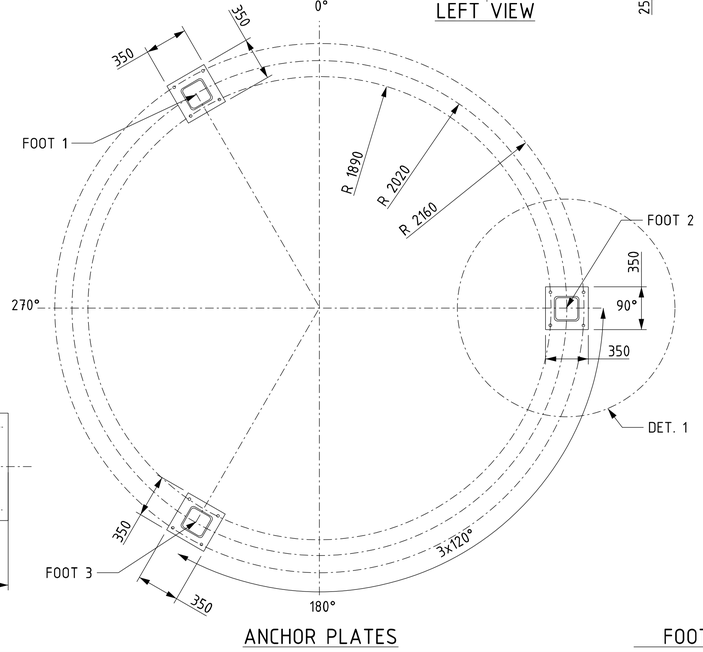


TOP VIEW

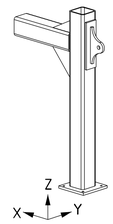
N23



DET. 1
1:10



ANCHOR PLATES



FOOTING COORDINATES

DESIGN DATA

EQUIPMENT TAG NOs. / NAME PLATE	3670-TNK-001
DESCRIPTION	CYANIDE DESTRUCTION PUMP TANK
DESIGN CODE	API 650
DESIGN TEMPERATURE	+0-60°C
WIND PRESSURE	90.97 kPa Absolute
OPERATING PRESSURE	ATM
WIND PRESSURE/ CODE	0.977 kPa/ SBC 301,2007
CORROSION ALLOWANCE: SHELL INSIDE	2 mm (6 THK. RUBBER LINING PROVIDED)
CORROSION ALLOWANCE: BOTTOM	2 mm (6 THK. RUBBER LINING PROVIDED)
TESTING	AS PER APPROVED Q.A.P
RADIOGRAPHY	AS PER APPROVED Q.A.P
JOINT EFFICIENCY	AS PER APPROVED Q.A.P
LOCATION	OUTDOOR
MAIN DIMENSIONS	4645mmx4460mmx6130mm
SHELL HEIGHT	2365mm
HEIGHT OF BASE	3130mm
CAPACITY: EFFECTIVE VOLUME	25m³
SEISMIC, WIND DESIGN /CODE	MD-216-1000-EG-GE-SPC-1001 /SBC 301, 2007

LOAD DATA

TOTAL WEIGHT (FULL)	60107 kg
TOTAL WEIGHT (EMPTY)	7817 kg
RUBBER LINING WEIGHT	340 kg
TANK WEIGHT	7474 kg
DETAIL WEIGHTS AND MATERIAL THICKNESSES ACCORDING TO VENDOR DETAIL ENGINEERING	

FOOTING LOAD TABLE

	DEAD LOAD			SEISMIC			WIND			LIVE LOAD			FLUID LOAD		
FOOT	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3
SHEAR FXY (kN)	0.01	0.02	0.01	6.44	0.21	6.44	8.3	0.3	8.3	0.00	0.00	0.00	0.07	0.07	0.07
AXIAL FZ (kN)	25.25	21.35	26.72	8.05	8.05	7.98	11.1	12.9	11.1	3.48	1.76	2.78	201.76	107.76	203.45
MAX MX (kNm)	0.00	0.00	0.01	2.52	0.35	2.52	3.9	0.6	3.9	0.00	0.00	0.00	0.03	0.00	0.03
MAX MY (kNm)	0.06	0.06	0.06	2.94	0.28	2.94	3.8	0.4	4.1	0.01	0.01	0.01	0.24	0.21	0.24
MAX MZ (kNm)	0.00	0.00	0.00	0.07	0.0	0.07	0.2	0.0	0.2	0.00	0.00	0.00	0.00	0.00	0.00

NOTE! Footing Loads: Dead, Live and Fluid Loads are unfactored. Seismic and Wind Loads are factored ACC to SBC301, EQ 2.4.1-5
NOTE! Moments and shear forces can act in any direction

MATERIAL OF CONSTRUCTION

PLATE - BOTTOM/SHELL/LIFTING LUGS	ASTM A284/ EN10025-2 S235JR
STRUCTURALS	ASTM A284/ EN10025-2 S235JR
NOZZLE	ASTM A284/ EN10025-2 S235JR
NOZZLE FLANGES	ASME B16.5 SO FF, ASTM A284
SURFACE TREATMENT	EN ISO 12944-5: EP240/4 - FE SA 2½
RUBBER LINING (INNER SURFACES)	NATURAL RUBBER LINER: HARDNESS ±40 ShA

NOZZLES TABLE

POS	SIZE	RATING	DIRECTION	PROJECTION	OFFSET	ELEVATION	STANDARD	SERVICE
N21	16"	150	260°	320	-	240	ASME B16.5	OUTLET
N22	16"	150	305°	320	-	240	ASME B16.5	OUTLET
N10	4"	150	235°	320	-	125	ASME B16.5	DRAIN
N11	24"	150	45°	320	-	3600	ASME B16.5	EMERGENCY OVERFLOW
K1	4"	150	270°	-	-	-	ASME B16.5	LEVEL INDICATOR/HOLD!

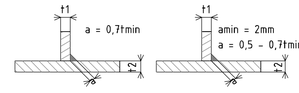
NOTE:
1. ALL DIMENSIONS ARE IN "mm" UNLESS OTHERWISE STATED.
2. ALL ELEVATION MEASUREMENTS ARE MEASURED FROM BOTTOM SIDE OF BASE PLATE.
3. SHELL NOZZLE ELEVATIONS ARE FROM BOTTOM FROM BOTTOM OF BASE PLATE TO CENTERLINE OF NOZZLE.
4. SHELL NOZZLE PROJECTIONS ARE MEASURED FROM OUTSIDE OF SHELL TO FLANGE FACE.
5. ALL NOZZLE LOCATION, ELEVATION AND ORIENTATION ARE INDICATIVE ONLY AND WILL BE FINALIZED DURING DETAIL DESIGN PHASE.

WELDING GENERAL NOTES

ALL WELDING SEAMS , MUST BE WELDED CONTINOUSLY UNLESS OTHERWISE SPECIFIED.

TYPICAL FILLET WELD THICKNESS:

ONE SIDE WELDING : GENERAL :



EXECUTE WELDING WORK ACCORDING TO DIN EN ISO 5817:
- FOR BUTT WELDS, QUALITY LEVEL C,
- FOR FILLET WELDS, QUALITY LEVEL B,

GENERAL TOLERANCE FOR WELDED CONSTRUTION ACCORDING TO EN ISO 13920 - AE.

GENERAL TOLERANCE acc. to ISO 2768 mH

MM	Machined MM	none Machined MM	Straightness Flatness MM
<= 6	± 0.1	± 0.5	0.008
> 6 - 30	± 0.2	± 1.0	0.020
> 30 - 120	± 0.3	± 1.5	0.040
> 120 - 400	± 0.5	± 2.0	0.080
> 400 - 1000	± 0.8	± 3.0	0.150
> 1000 - 2000	± 1.2	± 4.0	0.200
> 2000 - 4000	± 2.0	± 6.0	
> 4000	± 3.0	± 8.0	

CONCENTRIC and AXIAL RUN TOLERANCE 0.1 MM

SYMMETRIC TOLERANCE 0.5 MM

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A01	31-01-20	ISSUED FOR APPROVAL	KL	TV	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER 019045 450 M G 02441 004 | REVISION 0

OUTOTEC DRAWING NUMBER 900707_GAU01_RN12_MTA01_36703 | DOCUMENT ID OU100372788 | REVISION 0

DRAWN	KimLop	31-01-20
CHECKED	TatVie	31-01-20
ENGINEER	SouCha	31-01-20
APPROVED	RicLeh	31-01-20
EMPLOYER		
SCALE	1:25	A1

MANSOURAH - MASSARAH GOLD PROJECT
OUTLINE DRAWING
PRIMARY CYANIDE DESTRUCTION
3670-TNK-001 CYANIDE DESTRUCTION PUMP TANK

DRAWING NUMBER MD-216-3670-EG-ME-GAD-1007 | REV A01

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DRAWING NUMBER MD-216-3670-EG-ME-GAD-1010 REV A02

B 1013

C 1014

PREVAILING WIND DIRECTION

PLANT NORTH

32° N GEODETIC NORTH

25° W MECCA

S

E= 264339.101
N= 2506012.157

TOC EL +911.840 A LP

TOC EL +912.200 A HP

TOC EL +911.840 A LP

TOC EL +912.200 A HP

TOC EL +912.200 A HP

CANAL

TOS EL. +914.092

TOS EL. +914.170

TOS EL. +913.662

SLOPE 1:50

1717

3670-ANL-001
CYANIDE WAD ANALYZER

3670-001
HOLD

3670-REA-002
CYANIDE DESTRUCTION REACTOR

TOR EL +913.635

3670-REA-001
CYANIDE DESTRUCTION REACTOR

3670-REA-003
CYANIDE DESTRUCTION REACTOR

3670-003
HOLD

3670-PMP-001
CYANIDE DESTRUCTION TAILS PUMP

3670-004
HOLD

3670-PMP-002
CYANIDE DESTRUCTION TAILS PUMP

3740-001
HOLD

3740-PMP-001
SECONDARY CYANIDE
DESTRUCTION DISCHARGE PUMP

3740-002
HOLD

3740-PMP-002
SECONDARY CYANIDE
DESTRUCTION DISCHARGE PUMP

3740-003
HOLD

3740-PMP-101
SECONDARY CYANIDE
DESTRUCTION AREA SUMP PUMP

E= 264325.687
N= 2505958.141

TOC EL +911.840 A LP

TOS EL. +913.790

TOC EL +912.116 HP A

TOC EL +912.200 A HP

TOR EL +913.635

3740-REA-001
SECONDARY CYANIDE DESTRUCTION REACTOR

3670-TNK-001
CYANIDE DESTRUCTION PUMP TANK

SERVICE AREA

3670-PMP-101
CYANIDE DESTRUCTION AREA SUMP PUMP

3670-005
HOLD

A 1012

3670/F
3670/E
3670/D
3670/C
3670/B
3670/A

10000
6100
6100
10000
6500

38700

3670/1 3670/2 3670/3 3670/4 3670/5 3670/6 3670/7

5000 8000 7000 7000 8000 5000

40000

4000

1400 2850 5100 6000 1800 2050 2100 700 1250 250

35° 10°

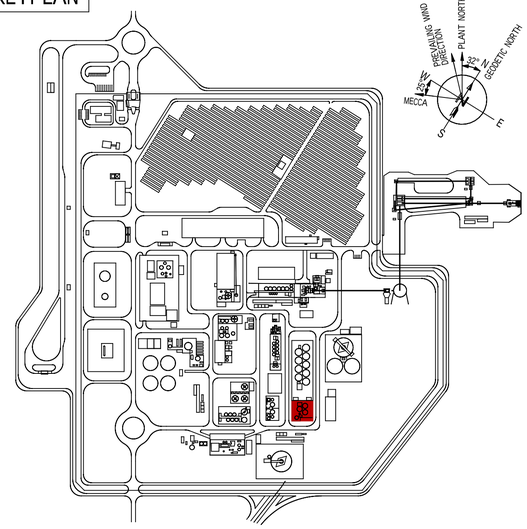
0 1 10m

PLAN AT EL.(+)912.200

CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
CONCRETE FLOOR
A — CONCRETE FLOOR WITH ALKALI / ACID RESISTANT COATING
CHECKERED PLATE
GRATING
F — GRATING - FRP
ACID RESISTANT BRICKLINING
BRICK WALL
CONCRETE WALL

1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METERS UNLESS OTHERWISE STATED.
2. FINISHED GROUND LEVEL (F.G.L) FOR PROCESS AREA 3670 = RL(+)912.200 m
3. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE. T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE. T.O.G. = TOP OF GRATING. HP = HIGHT POINT. LP = LOW POINT. T.O.R = TOP OF RAIL.
4. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3000-EG-GM-LST-1010	HOLD LIST								
MD-216-3670-EG-ME-GAD-1014	SECTION C-C								
MD-216-3670-EG-ME-GAD-1013	SECTION B-B								
MD-216-3670-EG-ME-GAD-1012	SECTION A-A	A02	27-01-2020	ISSUED FOR DESIGN	BR	TV	SC	RL	
MD-216-3670-EG-ME-GAD-1011	PLAN VIEW	A01	15-10-2019	ISSUED FOR APPROVAL	WK	TV	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER: 019045_450_M_G_02955_001 — REVISION: 1

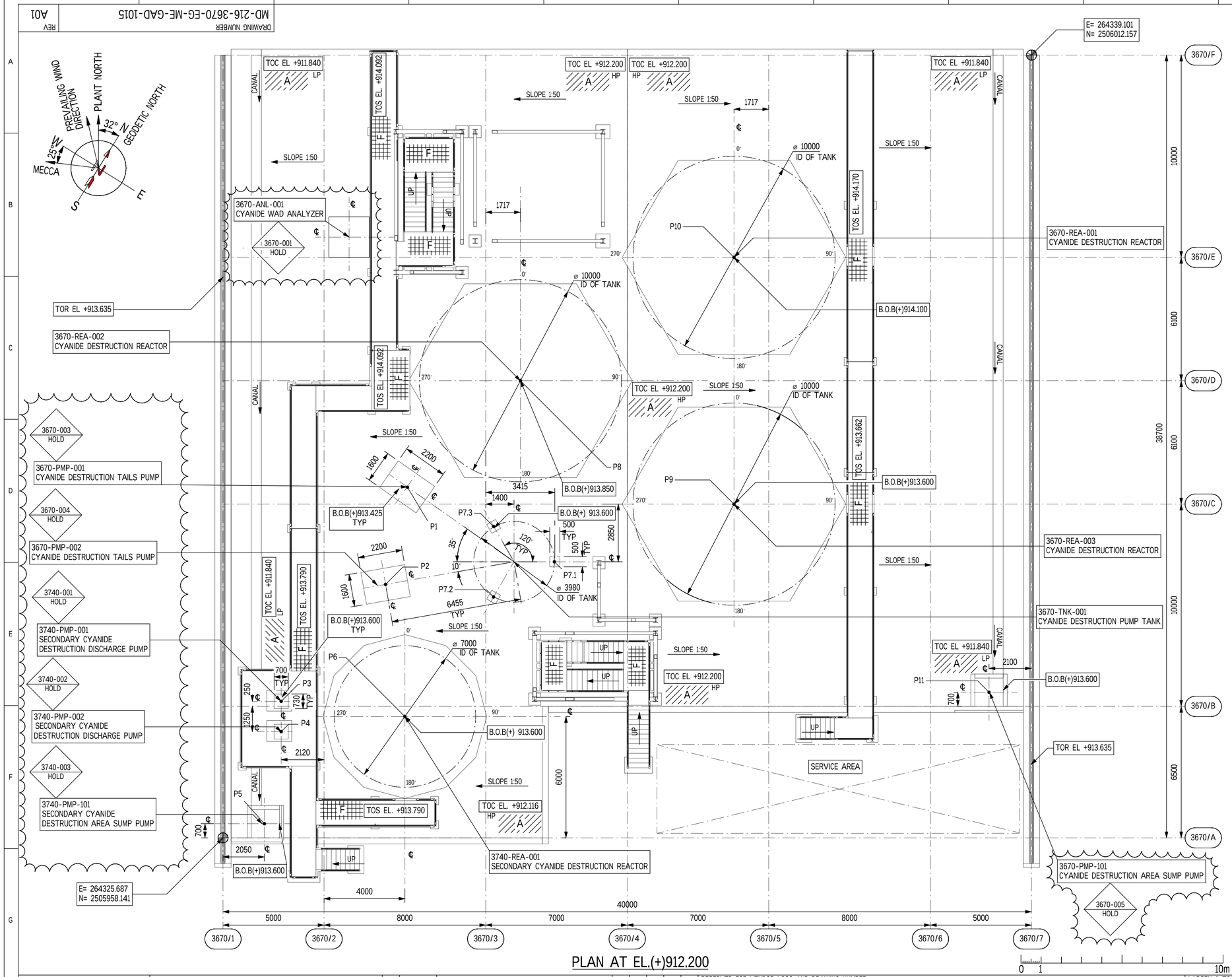
OUTOTEC DRAWING NUMBER: 900707 _GAU01 _RN12 _MTL01 _36701 — DOCUMENT ID: OU100664280 — REVISION: 1

DRAWN	WK	15-10-2019
CHECKED	TV	15-10-2019
ENGINEER	SC	15-10-2019
APPROVED	RL	15-10-2019
EMPLOYER		
SCALE	1:100	A1

MANSOURAH-MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
PRIMARY CYANIDE DESTRUCTION
PLAN AT EL. +912.200

DRAWING NUMBER: MD-216-3670-EG-ME-GAD-1010 — REV: A02

DRAWING NUMBER MD-216-3670-EG-ME-GAD-1015 REV A01



PLAN AT EL.(+)912.200

CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
CONCRETE FLOOR
CONCRETE FLOOR WITH ALKALI / ACID RESISTANT COATING
CHECKERED PLATE
GRATING
GRATING - FRP
ACID RESISTANT BRICKLINING
BRICK WALL
CONCRETE WALL

GENERAL NOTES, LEGEND

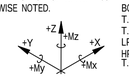
01 - ALL DIMENSION ARE IN MILLIMETER. 02 - ALL COORDINATE AND LEVELS ARE IN METER.

03 - LOADS ARE INDICATED IN KILONEWTON[kN]; KILONEWTONMETER[kNm] OR KILONEWTON/ SQUAREMETER [kN/m²] UNLESS OTHERWISE NOTED.

04 - ALL LOADS REFER TO TOP OF BEAM UNLESS OTHERWISE NOTED.

05 - T.O.S., B.O.S, B.O.B. = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
T.O.C. = TOP OF CONCRETE
T.O.R. = TOP OF RAIL
LP = LOW POINT
HP = HIGHT POINT
T.O.R = TOP OF RAIL.

DIRECTION OF FORCES AND MOMENTS



06 - THE LOADS SHOWN DO NOT INCLUDE :
- LOADS FROM AIR CONDITIONING EQUIPMENT
- ADDITIONAL LOADS FOR ERECTION PURPOSES

07 - FLOOR DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
OPERATING PLATFORM-GRATING 5 kN/m² GROUNDFLOOR NOT ACCESSIBLE TO TRUCKS 10 kN/m²
HEAVY STORAGE AREAS ACCESSIBLE BY VEHICLE OR CRANE 30 kN/m²
ELECTRICAL SWITCH GEAR AREAS 10 kN/m²

08 - ROOF DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
DUST 0.5 kN/m² ROOF NOT ACCESSED FOR MAINTENANCE 0.8 kN POINT LOAD

09 - FOR ITEM SUCH AS DUCTS, PIPES, CABLE TRAYS, ETC. NOT SHOWN SEPARATELY, AN ADDITIONAL VERTICAL LOAD OF 50 kN SHALL BE PROVIDED ON TOP OF EACH COLUMN.

10 - STATIC LOADS :
FOR STATIC LOADS CERTAIN EQUIPMENT DATA ARE TAKEN INTO CONSIDERATION, AFTER PLACEMENT OF PURCHASE ORDER, STATIC LOADS FOR CERTAIN EQUIPMENT HAVE TO BE RECHECKED ACCORDING TO ACTUAL WEIGHTS.

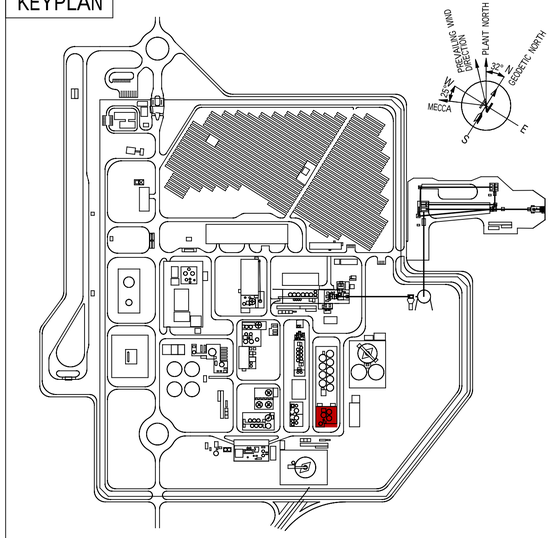
11 - DYNAMIC LOADS :
INFORMATION WITH REGARD TO DYNAMIC LOADS WHICH ARE NOT INDICATED AND HAVE TO BE REQUESTED WITH THE SUPPLIER OF THE RESPECTIVE EQUIPMENT, THIS INFORMATION IS REQUIRED FOR THE DETAIL ENGINEERING AS A ROUGH ESTIMATION FOR THE DYNAMIC LOADS. TAKE 1.5 TIMES OF THE STATIC LOADS , BASED ON THE INDICATIONS FOR USUAL EQUIPMENT SUPPLIERS.

12 - FINISHED GROUND LEVEL (F.G.L.) = RL(+)912.200 m.

13 - LOCATIONS AND SIZES OF PUMP FOUNDATION ARE TENTATIVE AND WILL BE SHOWN IN DETAIL DESIGN PHASE.

14 - ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3670-EG-ME-GAD-1017	LOAD TABLE								
MD-216-3670-EG-ME-GAD-1010	GENERAL ARRANGEMENT DRAWING								
MD-216-3000-EG-GM-LST-1010	HOLD LIST								
MD-216-3670-EG-ME-GAD-1016	PLAN VIEW	A01	31-01-2020	ISSUED FOR APPROVAL	BR	TV	SC	RL	

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LARSEN & TOUBRO DRAWING NUMBER	019045_450_M_D_03258_001	REVISION 0
OUTOTEC DRAWING NUMBER	900707 _GAU01 _RN12 _MLD14 _36701	DOCUMENT ID OU100667835 / REVISION 0

DRAWN	BR	31-01-2020	MANSOURAH-MASSARAH GOLD PROJECT
CHECKED	TV	31-01-2020	LOAD PLAN
ENGINEER	SC	31-01-2020	PRIMARY CYANIDE DESTRUCTION
APPROVED	RL	31-01-2020	PLAN AT EL. +912.200
EMPLOYER			
SCALE 1:100		A1	DRAWING NUMBER MD-216-3670-EG-ME-GAD-1015 / REV A01

ATTACHMENT 2 COMMENT RESOLUTION LOG SHEET / COMPLIANCE REPORT

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Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

PROJECT: MANSOURAH MASSARAH GOLD PROJECT **PROJECT NO.: 216**

STAGE: 4 - PROJECT EXECUTION. **REVIEW LOG DOC. NO. & REV.:**

ACONEX REF. NO. & DATE: BECHTEL-WTRAN-002345, 5 May 2020

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 5 May 2020	Comment by	EPC Response DATE: 8 May 2020	Status	Remarks
			General Comments (cc)				
MD-216-3670-EG-CV-CAL-1356-A01	3670	1	Please be more specific with the name of the document. It will be required in the future for searching, title should include the equipment name	Bechtel	Equipment name is incorporated in the document title.	Closed	